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1 Supplementary Materials

This document accompanies the manuscript “*Effects of a fixed-time food delivery on the behavior of domestic cats (*Felis catus*)*”. It contains the full statistical output for the analyses summarised in the Results section of the paper.

The supplement is organised in the same order as the paper:

1. Step counts
2. Activity patterns across the food-to-food interval
3. Space allocation (time near the feeder)
4. Water consumption

All tables in this supplement are regenerated from the raw data (`data/`) by running `SCRIPT_compile_stat_results.py`. The figures reproduced here are the same figures that appear in the main text.

2 Step Counts

Daily step counts were recorded throughout the study using a Fitbit Zip on each cat's collar. To test whether FT food delivery altered step counts beyond intrinsic baseline variation, we fit a per-cat OLS regression of daily step count on day using baseline observations only, then compared the distribution of residuals from that fit against the residuals obtained by applying the same model to the FT-interval days. A Kolmogorov–Smirnov two-sample test was used to test whether the two distributions could be drawn from the same underlying distribution. Figure 2 (reproduced below) shows the raw daily step counts and the fitted baseline trend for each cat.

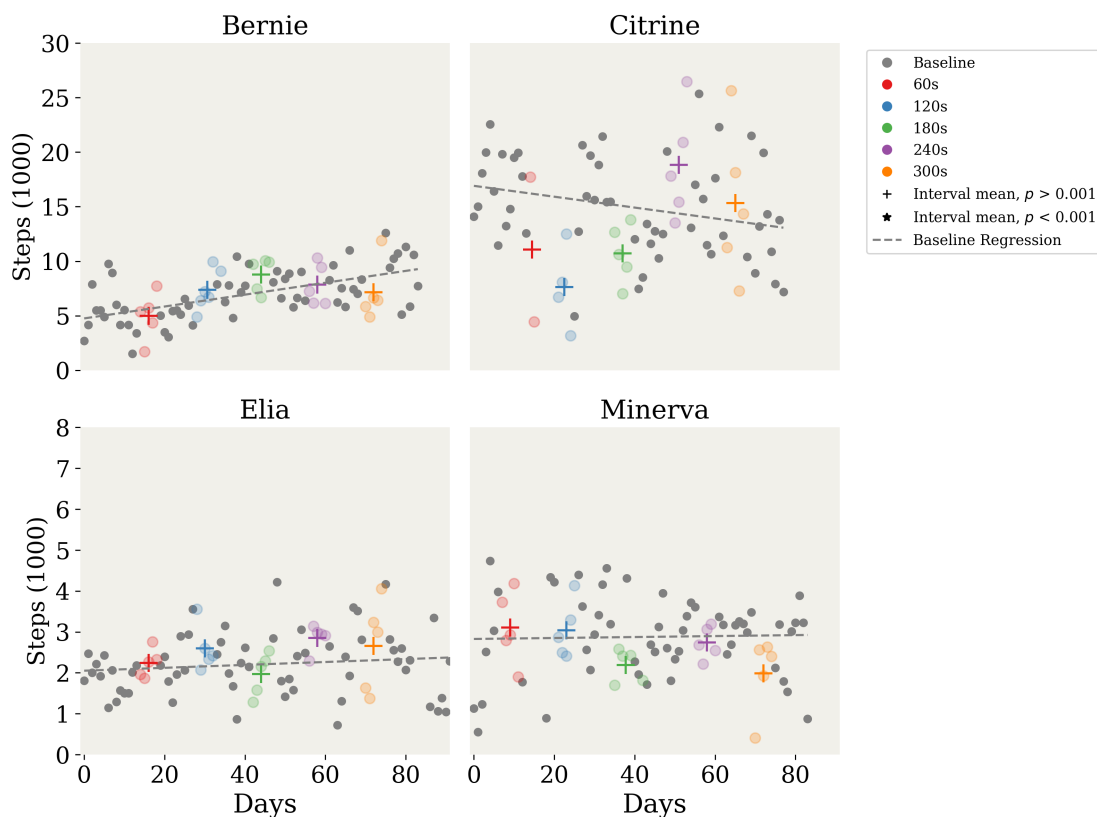


Figure 1: Figure 2. Daily step counts across baseline and FT intervals.

2.1 Baseline regression models

The tables below give the per-cat baseline OLS regressions of daily step count (in thousands) on day number, fit on baseline observations only. For each cat, the model is:

`Steps ~ Days`

The `Days` coefficient captures the intrinsic linear trend across the study period; the residuals from these fits are used in the Kolmogorov–Smirnov tests reported in the next subsection.

2.1.1 Bernie

OLS Regression Results

Dep. Variable:	steps	R-squared:	0.352
Model:	OLS	Adj. R-squared:	0.340

```

Method:          Least Squares    F-statistic:          30.41
Date:            Mon, 01 Jun 2026  Prob (F-statistic):       9.23e-07
Time:            18:24:07          Log-Likelihood:        -120.19
No. Observations: 58              AIC:                  244.4
Df Residuals:    56              BIC:                  248.5
Df Model:        1
Covariance Type: nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    4.7564      0.476     10.001     0.000      3.804      5.709
Days         0.0544      0.010      5.515     0.000      0.035      0.074
=====
Omnibus:            0.593    Durbin-Watson:           1.952
Prob(Omnibus):      0.743    Jarque-Bera (JB):        0.664
Skew:               0.223    Prob(JB):               0.717
Kurtosis:           2.724    Cond. No.                89.2
=====

```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

2.1.2 Citrine

OLS Regression Results

```

=====
Dep. Variable:      steps    R-squared:          0.072
Model:              OLS      Adj. R-squared:      0.054
Method:             Least Squares    F-statistic:        3.895
Date:               Mon, 01 Jun 2026  Prob (F-statistic):    0.0540
Time:               18:24:07          Log-Likelihood:      -150.25
No. Observations:   52          AIC:                  304.5
Df Residuals:       50          BIC:                  308.4
Df Model:           1
Covariance Type:    nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    16.9085      1.161     14.557     0.000     14.576     19.241
Days        -0.0498      0.025     -1.973     0.054     -0.101      0.001
=====
Omnibus:            0.695    Durbin-Watson:           1.574
Prob(Omnibus):      0.706    Jarque-Bera (JB):        0.522
Skew:               0.242    Prob(JB):               0.770
Kurtosis:           2.922    Cond. No.                86.9
=====

```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

2.1.3 Elia

OLS Regression Results

```

=====
Dep. Variable:      steps    R-squared:          0.017

```

```

Model:                      OLS      Adj. R-squared:      0.001
Method:                     Least Squares      F-statistic:      1.054
Date:                       Mon, 01 Jun 2026      Prob (F-statistic):      0.309
Time:                       18:24:07      Log-Likelihood:      -70.141
No. Observations:          62      AIC:      144.3
Df Residuals:              60      BIC:      148.5
Df Model:                  1
Covariance Type:           nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept      2.0513      0.179      11.491      0.000      1.694      2.408
Days           0.0035      0.003       1.027      0.309     -0.003      0.010
=====
Omnibus:              1.439      Durbin-Watson:      1.441
Prob(Omnibus):        0.487      Jarque-Bera (JB):      0.955
Skew:                 0.297      Prob(JB):      0.620
Kurtosis:             3.133      Cond. No.      95.2
=====

```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

2.1.4 Minerva

OLS Regression Results

```

=====
Dep. Variable:          steps      R-squared:      0.001
Model:                  OLS      Adj. R-squared:      -0.018
Method:                 Least Squares      F-statistic:      0.04356
Date:                   Mon, 01 Jun 2026      Prob (F-statistic):      0.835
Time:                   18:24:07      Log-Likelihood:      -76.059
No. Observations:      54      AIC:      156.1
Df Residuals:          52      BIC:      160.1
Df Model:              1
Covariance Type:       nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept      2.8276      0.286       9.881      0.000      2.253      3.402
Days           0.0012      0.006       0.209      0.835     -0.010      0.012
=====
Omnibus:              1.117      Durbin-Watson:      1.136
Prob(Omnibus):        0.572      Jarque-Bera (JB):      1.170
Skew:                 -0.292      Prob(JB):      0.557
Kurtosis:             2.578      Cond. No.      106.
=====

```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

2.2 Kolmogorov–Smirnov tests of FT vs. baseline residual distributions

Each row compares the distribution of residuals from the corresponding cat’s baseline regression against the distribution of residuals obtained by applying that same baseline model to the FT interval named in the comparison column. A p -value below 0.001 indicates that the FT-interval residuals differ significantly from baseline residuals — i.e., that the intervention altered step counts beyond what would be expected from intrinsic baseline variation.

Subject	Comparison	KS statistic	p-value
Bernie	interval 60s vs Baseline	ks = 0.283	p = 0.763
Bernie	interval 120s vs Baseline	ks = 0.351	p = 0.415
Bernie	interval 180s vs Baseline	ks = 0.497	p = 0.147
Bernie	interval 240s vs Baseline	ks = 0.210	p = 0.955
Bernie	interval 300s vs Baseline	ks = 0.645	p = 0.023
Citrine	interval 60s vs Baseline	ks = 0.500	p = 0.563
Citrine	interval 120s vs Baseline	ks = 0.808	p = 0.005
Citrine	interval 180s vs Baseline	ks = 0.519	p = 0.119
Citrine	interval 240s vs Baseline	ks = 0.481	p = 0.178
Citrine	interval 300s vs Baseline	ks = 0.246	p = 0.884
Elia	interval 60s vs Baseline	ks = 0.355	p = 0.499
Elia	interval 120s vs Baseline	ks = 0.452	p = 0.228
Elia	interval 180s vs Baseline	ks = 0.290	p = 0.739
Elia	interval 240s vs Baseline	ks = 0.639	p = 0.025
Elia	interval 300s vs Baseline	ks = 0.439	p = 0.250
Minerva	interval 60s vs Baseline	ks = 0.226	p = 0.929
Minerva	interval 120s vs Baseline	ks = 0.296	p = 0.728
Minerva	interval 180s vs Baseline	ks = 0.630	p = 0.031
Minerva	interval 240s vs Baseline	ks = 0.389	p = 0.396
Minerva	interval 300s vs Baseline	ks = 0.630	p = 0.031

3 Activity Patterns Across the Food-to-Food Interval

Activity was scored from video using a 15-second instantaneous time-sampling method; at each sample point the cat was classified as active (any movement or interaction) or inactive (resting or pausing). To assess how activity was allocated across each food-to-food interval, we fit two logistic regressions (`smf.logit`) per FT interval, pooling data across cats: one for the first two-thirds of the interval and one for the final third. Each model used `TimeSinceFeeding` and `C(Subject)` as predictors, and we tested whether the slope on `TimeSinceFeeding` differed significantly from zero. For the 60s interval, which contains only four sample points, the two-thirds split falls between the second and third points, so the two segments are effectively halves rather than thirds. Figure 3 (reproduced below) shows the average proportion of intervals marked as active across the food-to-food interval. Open circles mark data points in the first two-thirds of each interval, and filled circles mark those in the final one-third, corresponding to the two segments fit by the piecewise models below.

3.1 Logistic models per FT interval (first 2/3 and final 1/3)

For each FT interval below, the first table gives the logit fit for the first segment and the second table for the final segment. A positive slope on `TimeSinceFeeding` indicates that the probability of being active rises across the segment; a negative slope indicates it falls. The p -value on the `TimeSinceFeeding` row tests whether the slope differs significantly from zero.

3.1.1 60s interval

First segment (first 2/3 of interval):

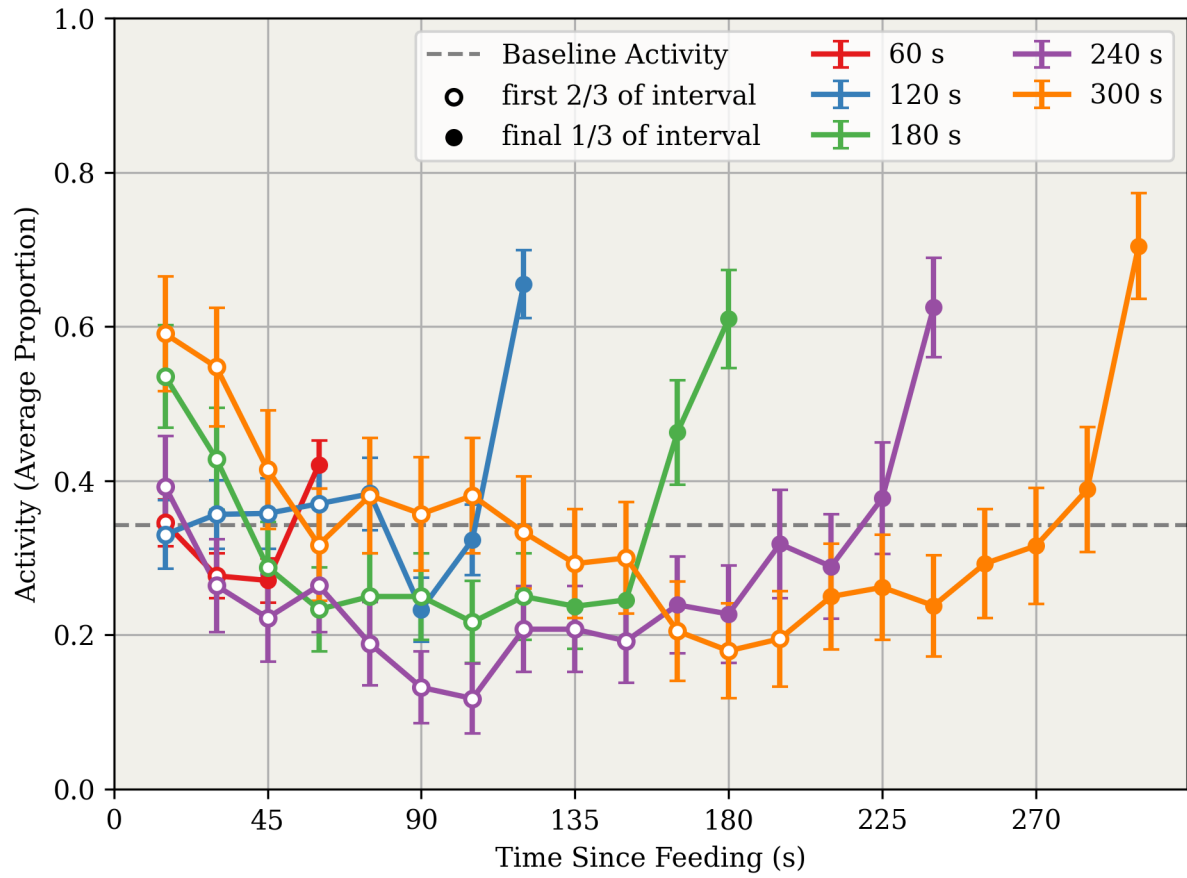


Figure 2: Figure 3. Average proportion of intervals marked as active across the food-to-food interval.

Logit Regression Results

=====						
Dep. Variable:	Active	No. Observations:	488			
Model:	Logit	Df Residuals:	484			
Method:	MLE	Df Model:	3			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.04769			
Time:	18:24:07	Log-Likelihood:	-288.26			
converged:	True	LL-Null:	-302.69			
Covariance Type:	nonrobust	LLR p-value:	2.387e-06			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	0.4877	0.351	1.390	0.164	-0.200	1.175
Subject = Elia	-1.1217	0.247	-4.533	0.000	-1.607	-0.637
Subject = Minerva	-1.0771	0.248	-4.341	0.000	-1.563	-0.591
TimeSinceFeeding	-0.0231	0.013	-1.708	0.088	-0.050	0.003
=====						

Second segment (final 1/3 of interval):

Logit Regression Results

=====						
Dep. Variable:	Active	No. Observations:	487			
Model:	Logit	Df Residuals:	483			
Method:	MLE	Df Model:	3			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.03339			
Time:	18:24:07	Log-Likelihood:	-303.90			
converged:	True	LL-Null:	-314.40			
Covariance Type:	nonrobust	LLR p-value:	0.0001056			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	-2.6132	0.714	-3.660	0.000	-4.012	-1.214
Subject = Elia	-0.5354	0.241	-2.219	0.026	-1.008	-0.063
Subject = Minerva	-0.7026	0.245	-2.864	0.004	-1.183	-0.222
TimeSinceFeeding	0.0458	0.013	3.498	0.000	0.020	0.071
=====						

3.1.2 120s interval

First segment (first 2/3 of interval):

Logit Regression Results

=====						
Dep. Variable:	Active	No. Observations:	551			
Model:	Logit	Df Residuals:	546			
Method:	MLE	Df Model:	4			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1328			
Time:	18:24:07	Log-Likelihood:	-312.03			
converged:	True	LL-Null:	-359.83			
Covariance Type:	nonrobust	LLR p-value:	8.516e-20			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	0.6358	0.275	2.313	0.021	0.097	1.175
Subject = Citrine	-1.3530	0.268	-5.052	0.000	-1.878	-0.828

Subject = Elia	-2.0174	0.252	-8.020	0.000	-2.510	-1.524
Subject = Minerva	-2.2682	0.305	-7.438	0.000	-2.866	-1.670
TimeSinceFeeding	0.0027	0.005	0.575	0.566	-0.006	0.012

Second segment (final 1/3 of interval):

Logit Regression Results

Dep. Variable:	Active	No. Observations:	324
Model:	Logit	Df Residuals:	319
Method:	MLE	Df Model:	4
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1456
Time:	18:24:07	Log-Likelihood:	-187.73
converged:	True	LL-Null:	-219.72
Covariance Type:	nonrobust	LLR p-value:	4.252e-13

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-6.8073	1.153	-5.904	0.000	-9.067	-4.547
Subject = Citrine	-1.2505	0.379	-3.303	0.001	-1.993	-0.508
Subject = Elia	-1.3703	0.333	-4.118	0.000	-2.022	-0.718
Subject = Minerva	-0.5576	0.362	-1.540	0.124	-1.267	0.152
TimeSinceFeeding	0.0683	0.011	6.298	0.000	0.047	0.090

3.1.3 180s interval

First segment (first 2/3 of interval):

Logit Regression Results

Dep. Variable:	Active	No. Observations:	471
Model:	Logit	Df Residuals:	466
Method:	MLE	Df Model:	4
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1733
Time:	18:24:07	Log-Likelihood:	-239.05
converged:	True	LL-Null:	-289.14
Covariance Type:	nonrobust	LLR p-value:	8.938e-21

	coef	std err	z	P> z	[0.025	0.975]
Intercept	1.3888	0.304	4.575	0.000	0.794	1.984
Subject = Citrine	-1.1504	0.304	-3.781	0.000	-1.747	-0.554
Subject = Elia	-2.1627	0.317	-6.814	0.000	-2.785	-1.541
Subject = Minerva	-2.2866	0.328	-6.972	0.000	-2.929	-1.644
TimeSinceFeeding	-0.0160	0.003	-4.637	0.000	-0.023	-0.009

Second segment (final 1/3 of interval):

Logit Regression Results

Dep. Variable:	Active	No. Observations:	229
Model:	Logit	Df Residuals:	224
Method:	MLE	Df Model:	4
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1054


```

Time:                18:24:07   Log-Likelihood:        -136.87
converged:           True       LL-Null:                -153.00
Covariance Type:    nonrobust   LLR p-value:         1.691e-06
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-6.4208	1.431	-4.488	0.000	-9.225	-3.617
Subject = Citrine	-1.0766	0.442	-2.433	0.015	-1.944	-0.209
Subject = Elia	-0.9195	0.384	-2.392	0.017	-1.673	-0.166
Subject = Minerva	-0.8926	0.390	-2.290	0.022	-1.656	-0.129
TimeSinceFeeding	0.0416	0.009	4.627	0.000	0.024	0.059

```

=====

```

3.1.4 240s interval

First segment (first 2/3 of interval):

```

                        Logit Regression Results
=====
Dep. Variable:          Active   No. Observations:          531
Model:                  Logit   Df Residuals:              526
Method:                 MLE     Df Model:                  4
Date:                  Mon, 01 Jun 2026   Pseudo R-squ.:            0.1936
Time:                  18:24:07   Log-Likelihood:           -225.81
converged:             True     LL-Null:                  -280.02
Covariance Type:      nonrobust   LLR p-value:             1.589e-22
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.6110	0.278	2.199	0.028	0.066	1.156
Subject = Citrine	-2.4003	0.373	-6.436	0.000	-3.131	-1.669
Subject = Elia	-1.4499	0.268	-5.404	0.000	-1.976	-0.924
Subject = Minerva	-3.1442	0.488	-6.437	0.000	-4.102	-2.187
TimeSinceFeeding	-0.0076	0.003	-2.756	0.006	-0.013	-0.002

```

=====

```

Second segment (final 1/3 of interval):

```

                        Logit Regression Results
=====
Dep. Variable:          Active   No. Observations:          280
Model:                  Logit   Df Residuals:              275
Method:                 MLE     Df Model:                  4
Date:                  Mon, 01 Jun 2026   Pseudo R-squ.:            0.1746
Time:                  18:24:07   Log-Likelihood:           -150.62
converged:             True     LL-Null:                  -182.49
Covariance Type:      nonrobust   LLR p-value:             4.746e-13
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-4.8439	1.168	-4.148	0.000	-7.133	-2.555
Subject = Citrine	-1.0072	0.363	-2.776	0.006	-1.718	-0.296
Subject = Elia	-2.2305	0.402	-5.548	0.000	-3.019	-1.443
Subject = Minerva	-1.9570	0.419	-4.675	0.000	-2.777	-1.137
TimeSinceFeeding	0.0261	0.006	4.576	0.000	0.015	0.037

```

=====

```

3.1.5 300s interval

First segment (first 2/3 of interval):

Logit Regression Results						
Dep. Variable:	Active	No. Observations:	536			
Model:	Logit	Df Residuals:	531			
Method:	MLE	Df Model:	4			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.09096			
Time:	18:24:07	Log-Likelihood:	-315.13			
converged:	True	LL-Null:	-346.66			
Covariance Type:	nonrobust	LLR p-value:	6.586e-13			
	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.8772	0.248	3.533	0.000	0.391	1.364
Subject = Citrine	-0.4818	0.263	-1.831	0.067	-0.998	0.034
Subject = Elia	-0.5221	0.242	-2.161	0.031	-0.996	-0.048
Subject = Minerva	-1.6867	0.312	-5.398	0.000	-2.299	-1.074
TimeSinceFeeding	-0.0094	0.002	-5.316	0.000	-0.013	-0.006

Second segment (final 1/3 of interval):

Logit Regression Results						
Dep. Variable:	Active	No. Observations:	283			
Model:	Logit	Df Residuals:	278			
Method:	MLE	Df Model:	4			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1142			
Time:	18:24:07	Log-Likelihood:	-162.82			
converged:	True	LL-Null:	-183.81			
Covariance Type:	nonrobust	LLR p-value:	1.690e-08			
	coef	std err	z	P> z	[0.025	0.975]
Intercept	-5.2401	1.198	-4.375	0.000	-7.588	-2.892
Subject = Citrine	-0.9090	0.372	-2.442	0.015	-1.638	-0.179
Subject = Elia	-0.7684	0.342	-2.249	0.025	-1.438	-0.099
Subject = Minerva	-1.8594	0.435	-4.279	0.000	-2.711	-1.008
TimeSinceFeeding	0.0210	0.005	4.524	0.000	0.012	0.030

4 Location (Time in Feeder Area)

At each 15-second sample point the cat's location was scored using a six-zone system covering the run, plus several non-zone categories (Shelf, Fence, Litter Box, Out of View; see `Library/AnalysisBehavior.read_data` for the full mapping). For analysis the location was collapsed into a binary `AtFeeder` outcome — Feeder Area vs. Non-Feeder; the exact zone-to-category mapping is in the `loc_cat` lambda of `read_data`. Three logistic models were fit on this outcome to characterise how proximity to the feeder varied across baseline and the FT conditions.

Figure 4 (reproduced below) shows the average proportion of intervals spent in the feeder area across the food-to-food interval. Open circles mark data points in the first two-thirds of each interval, and filled circles mark those in the final one-third, corresponding to the two segments fit by the piecewise models below.

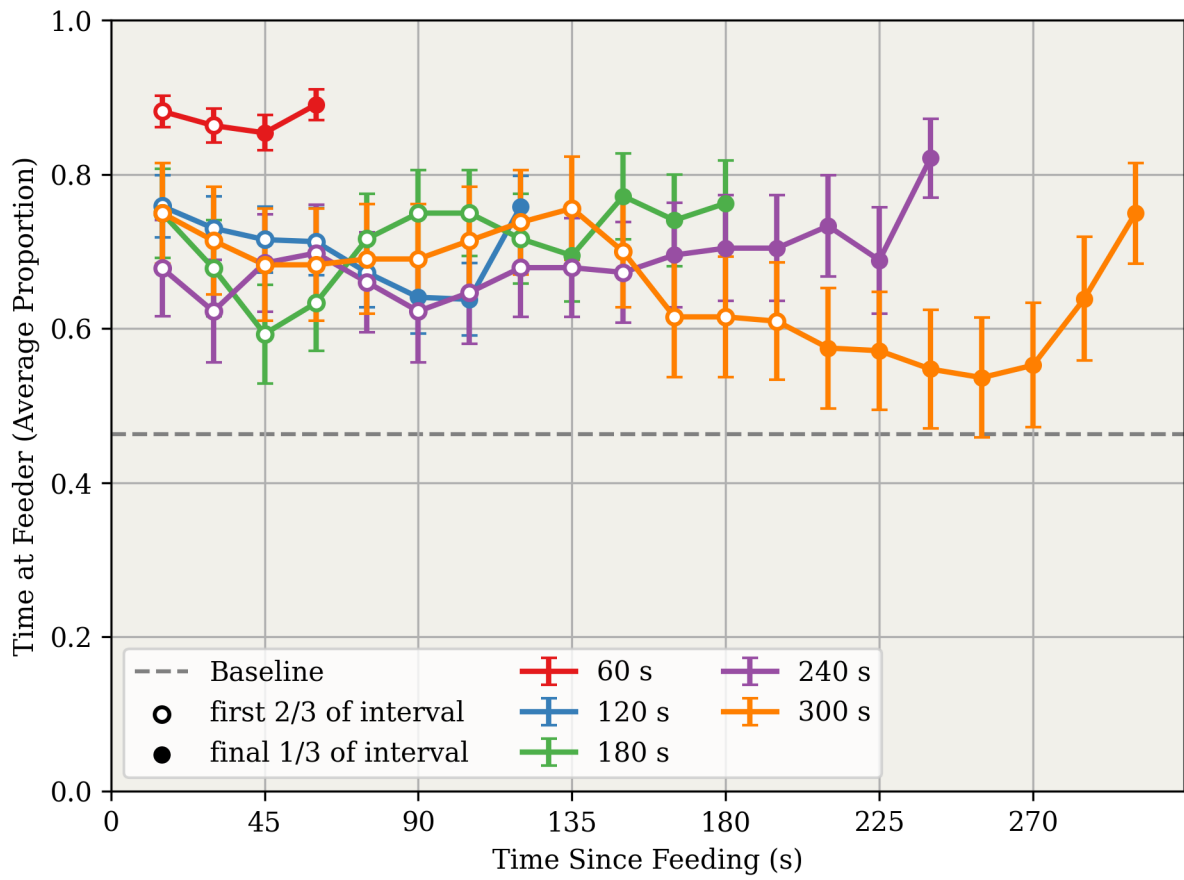


Figure 3: Figure 4. Average proportion of intervals spent in the feeder area.

4.1 Effect of FT interval (categorical, baseline as reference)

A logistic regression was fit on all observations (baseline plus all five FT intervals), pooling data across cats. FT interval was encoded as a categorical predictor with baseline as the reference level; subject was included as a fixed effect. The model is:

`AtFeeder ~ C(Subject) + C(Feeder_Interval) + TimeSinceFeeding`

Each `Feeder_Interval = N` coefficient tests whether time in the feeder area during that interval differs significantly from baseline.

Logit Regression Results						
=====						
Dep. Variable:	AtFeeder	No. Observations:	6479			
Model:	Logit	Df Residuals:	6469			
Method:	MLE	Df Model:	9			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1329			
Time:	18:24:07	Log-Likelihood:	-3679.8			
converged:	True	LL-Null:	-4244.1			
Covariance Type:	nonrobust	LLR p-value:	3.310e-237			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	-0.5430	0.056	-9.626	0.000	-0.654	-0.432
Subject = Citrine	-0.2339	0.076	-3.089	0.002	-0.382	-0.085
Subject = Elia	1.0726	0.079	13.653	0.000	0.919	1.227
Subject = Minerva	1.2707	0.083	15.334	0.000	1.108	1.433
Feeder_Interval = 60	1.8377	0.114	16.121	0.000	1.614	2.061
Feeder_Interval = 120	0.9389	0.098	9.541	0.000	0.746	1.132
Feeder_Interval = 180	1.0078	0.115	8.796	0.000	0.783	1.232
Feeder_Interval = 240	0.9202	0.119	7.747	0.000	0.687	1.153
Feeder_Interval = 300	0.7657	0.131	5.838	0.000	0.509	1.023
TimeSinceFeeding	-0.0003	0.001	-0.463	0.644	-0.001	0.001
=====						

4.2 Effect of interval length on time at feeder (continuous, intervention only)

A second logistic regression was fit on the intervention observations only (baseline excluded), treating FT interval length as a continuous predictor. This tests whether longer FT intervals are associated with systematically more or less time in the feeder area. The model is:

`AtFeeder ~ C(Subject) + Feeder_Interval + TimeSinceFeeding`

A significant negative slope on `Feeder_Interval` indicates that as interval length increases, cats spend less time near the feeder.

Logit Regression Results						
=====						
Dep. Variable:	AtFeeder	No. Observations:	4180			
Model:	Logit	Df Residuals:	4174			
Method:	MLE	Df Model:	5			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1881			
Time:	18:24:07	Log-Likelihood:	-1970.0			
converged:	True	LL-Null:	-2426.4			
Covariance Type:	nonrobust	LLR p-value:	4.694e-195			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	0.4190	0.120	3.490	0.000	0.184	0.654
Subject = Citrine	0.2379	0.097	2.463	0.014	0.049	0.427
Subject = Elia	2.2132	0.114	19.440	0.000	1.990	2.436
Subject = Minerva	2.4406	0.131	18.631	0.000	2.184	2.697
Feeder_Interval	-0.0025	0.001	-4.391	0.000	-0.004	-0.001
TimeSinceFeeding	-0.0003	0.001	-0.478	0.633	-0.002	0.001

4.3 Logistic models per FT interval (first 2/3 and final 1/3)

For each FT interval the data was split into the first two-thirds of the food-to-food interval and the final third, and a logistic regression was fit separately to each segment, using `TimeSinceFeeding` and `C(Subject)` as predictors. For the 60s interval, which contains only four sample points, the two-thirds split falls between the second and third points so the two segments are effectively halves. For the 240s final-third model, Minerva was excluded because she was at the feeder in all 57 of her observations in that segment, which caused the logit fit to diverge; her exclusion did not change the `TimeSinceFeeding` slope or p -value to four decimal places.

For each interval below, the first table gives the logit fit for the first segment and the second table for the final segment. A positive slope on `TimeSinceFeeding` indicates that the probability of being in the feeder area rises across the segment; a negative slope indicates it falls.

4.3.1 60s interval

First segment (first 2/3 of interval):

Logit Regression Results						
Dep. Variable:	AtFeeder	No. Observations:	488			
Model:	Logit	Df Residuals:	484			
Method:	MLE	Df Model:	3			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.04302			
Time:	18:24:08	Log-Likelihood:	-177.81			
converged:	True	LL-Null:	-185.80			
Covariance Type:	nonrobust	LLR p-value:	0.001141			
	coef	std err	z	P> z	[0.025	0.975]
Intercept	1.4565	0.471	3.094	0.002	0.534	2.379
Subject = Elia	1.0688	0.329	3.248	0.001	0.424	1.714
Subject = Minerva	1.1694	0.341	3.429	0.001	0.501	1.838
TimeSinceFeeding	-0.0111	0.018	-0.603	0.547	-0.047	0.025

Second segment (final 1/3 of interval):

Logit Regression Results						
Dep. Variable:	AtFeeder	No. Observations:	487			
Model:	Logit	Df Residuals:	483			
Method:	MLE	Df Model:	3			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1249			
Time:	18:24:08	Log-Likelihood:	-162.47			
converged:	True	LL-Null:	-185.66			
Covariance Type:	nonrobust	LLR p-value:	4.669e-10			

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.4107	1.021	-0.402	0.688	-2.412	1.591
Subject = Elia	2.2374	0.410	5.457	0.000	1.434	3.041
Subject = Minerva	1.5558	0.332	4.689	0.000	0.905	2.206
TimeSinceFeeding	0.0238	0.019	1.237	0.216	-0.014	0.062

4.3.2 120s interval

First segment (first 2/3 of interval):

Logit Regression Results						
Dep. Variable:	AtFeeder	No. Observations:	551			
Model:	Logit	Df Residuals:	546			
Method:	MLE	Df Model:	4			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1108			
Time:	18:24:08	Log-Likelihood:	-291.13			
converged:	True	LL-Null:	-327.39			
Covariance Type:	nonrobust	LLR p-value:	6.637e-15			
	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.5298	0.277	1.915	0.055	-0.012	1.072
Subject = Citrine	0.0713	0.257	0.278	0.781	-0.432	0.574
Subject = Elia	1.0073	0.244	4.122	0.000	0.528	1.486
Subject = Minerva	2.7981	0.488	5.737	0.000	1.842	3.754
TimeSinceFeeding	-0.0061	0.005	-1.273	0.203	-0.015	0.003

Second segment (final 1/3 of interval):

Logit Regression Results						
Dep. Variable:	AtFeeder	No. Observations:	324			
Model:	Logit	Df Residuals:	319			
Method:	MLE	Df Model:	4			
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1681			
Time:	18:24:08	Log-Likelihood:	-168.54			
converged:	True	LL-Null:	-202.59			
Covariance Type:	nonrobust	LLR p-value:	5.718e-14			
	coef	std err	z	P> z	[0.025	0.975]
Intercept	-2.7080	1.177	-2.301	0.021	-5.015	-0.402
Subject = Citrine	0.4246	0.339	1.253	0.210	-0.240	1.089
Subject = Elia	1.7337	0.330	5.254	0.000	1.087	2.380
Subject = Minerva	3.0733	0.565	5.437	0.000	1.965	4.181
TimeSinceFeeding	0.0228	0.011	2.095	0.036	0.001	0.044

4.3.3 180s interval

First segment (first 2/3 of interval):

Logit Regression Results

```

=====
Dep. Variable:          AtFeeder    No. Observations:          471
Model:                  Logit       Df Residuals:              466
Method:                MLE         Df Model:                  4
Date:                  Mon, 01 Jun 2026   Pseudo R-squ.:            0.3545
Time:                  18:24:08    Log-Likelihood:           -186.10
converged:              True        LL-Null:                  -288.31
Covariance Type:       nonrobust    LLR p-value:              4.206e-43
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.3559	0.314	-1.132	0.258	-0.972	0.260
Subject = Citrine	-1.1455	0.307	-3.734	0.000	-1.747	-0.544
Subject = Elia	3.1282	0.487	6.423	0.000	2.174	4.083
Subject = Minerva	3.3490	0.536	6.250	0.000	2.299	4.399
TimeSinceFeeding	0.0058	0.004	1.525	0.127	-0.002	0.013

```

=====

```

Second segment (final 1/3 of interval):

Logit Regression Results

```

=====
Dep. Variable:          AtFeeder    No. Observations:          229
Model:                  Logit       Df Residuals:              224
Method:                MLE         Df Model:                  4
Date:                  Mon, 01 Jun 2026   Pseudo R-squ.:            0.2429
Time:                  18:24:08    Log-Likelihood:           -98.925
converged:              True        LL-Null:                  -130.66
Covariance Type:       nonrobust    LLR p-value:              5.395e-13
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.4151	1.653	-0.856	0.392	-4.654	1.824
Subject = Citrine	0.7878	0.408	1.930	0.054	-0.012	1.588
Subject = Elia	3.1168	0.639	4.881	0.000	1.865	4.368
Subject = Minerva	3.0679	0.640	4.797	0.000	1.814	4.321
TimeSinceFeeding	0.0079	0.010	0.764	0.445	-0.012	0.028

```

=====

```

4.3.4 240s interval

First segment (first 2/3 of interval):

Logit Regression Results

```

=====
Dep. Variable:          AtFeeder    No. Observations:          531
Model:                  Logit       Df Residuals:              526
Method:                MLE         Df Model:                  4
Date:                  Mon, 01 Jun 2026   Pseudo R-squ.:            0.2789
Time:                  18:24:08    Log-Likelihood:           -244.24
converged:              True        LL-Null:                  -338.68
Covariance Type:       nonrobust    LLR p-value:              9.200e-40
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.6434	0.269	-2.391	0.017	-1.171	-0.116

Subject = Citrine	0.7352	0.255	2.884	0.004	0.236	1.235
Subject = Elia	2.3926	0.287	8.349	0.000	1.831	2.954
Subject = Minerva	5.4468	1.020	5.343	0.000	3.449	7.445
TimeSinceFeeding	-0.0005	0.003	-0.193	0.847	-0.005	0.005

Second segment (final 1/3 of interval):

Logit Regression Results

Dep. Variable:	AtFeeder	No. Observations:	223
Model:	Logit	Df Residuals:	219
Method:	MLE	Df Model:	3
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1579
Time:	18:24:08	Log-Likelihood:	-120.48
converged:	True	LL-Null:	-143.07
Covariance Type:	nonrobust	LLR p-value:	8.488e-10

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.7117	1.239	-1.381	0.167	-4.141	0.717
Subject = Citrine	0.5725	0.340	1.685	0.092	-0.093	1.239
Subject = Elia	2.6130	0.480	5.443	0.000	1.672	3.554
TimeSinceFeeding	0.0077	0.006	1.287	0.198	-0.004	0.019

4.3.5 300s interval

First segment (first 2/3 of interval):

Logit Regression Results

Dep. Variable:	AtFeeder	No. Observations:	536
Model:	Logit	Df Residuals:	531
Method:	MLE	Df Model:	4
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.1970
Time:	18:24:08	Log-Likelihood:	-266.37
converged:	True	LL-Null:	-331.71
Covariance Type:	nonrobust	LLR p-value:	2.773e-27

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.2237	0.255	0.876	0.381	-0.277	0.724
Subject = Citrine	0.0965	0.251	0.385	0.700	-0.395	0.588
Subject = Elia	3.2052	0.421	7.611	0.000	2.380	4.031
Subject = Minerva	1.5448	0.286	5.400	0.000	0.984	2.105
TimeSinceFeeding	-0.0033	0.002	-1.762	0.078	-0.007	0.000

Second segment (final 1/3 of interval):

Logit Regression Results

Dep. Variable:	AtFeeder	No. Observations:	283
Model:	Logit	Df Residuals:	278
Method:	MLE	Df Model:	4
Date:	Mon, 01 Jun 2026	Pseudo R-squ.:	0.3280

Time:	18:24:08	Log-Likelihood:	-128.20			
converged:	True	LL-Null:	-190.78			
Covariance Type:	nonrobust	LLR p-value:	4.194e-26			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	-3.1388	1.340	-2.342	0.019	-5.765	-0.512
Subject = Citrine	-1.4327	0.412	-3.480	0.001	-2.240	-0.626
Subject = Elia	3.5266	0.636	5.543	0.000	2.280	4.773
Subject = Minerva	1.0834	0.370	2.929	0.003	0.358	1.808
TimeSinceFeeding	0.0116	0.005	2.241	0.025	0.001	0.022
=====						

5 Water Consumption

Water consumption was recorded during each experimental session (the period during which food was delivered on intervention days, or an equivalent fixed window on baseline days). The analysis reported here uses the in-session column (`session` in `data/SIP_data.csv`); a separate continuation period for Citrine and Elia (“phase 2”) is computed by `AnalysisDrink.read_data` but not used in the paper. To test whether FT food delivery altered in-session water consumption beyond intrinsic baseline variation, we fit a per-cat OLS regression of session water consumption (ml) on day using baseline observations only, then compared the distribution of residuals from that fit against the residuals obtained by applying the same model to the FT-interval days. A Kolmogorov–Smirnov two-sample test was used to test whether the two distributions could be drawn from the same underlying distribution. Figure 5 (reproduced below) shows the raw daily session consumption and the fitted baseline trend for each cat.

5.1 Baseline regression models (phase 1, in-session consumption)

The tables below give the per-cat baseline OLS regressions of in-session water consumption (ml) on day number, fit on baseline observations only. The model is:

`Session ~ Days`

The `Days` coefficient captures the intrinsic linear trend across the study period; the residuals from these fits are used in the Kolmogorov–Smirnov tests reported in the next subsection.

5.1.1 Bernie

OLS Regression Results

Dep. Variable:	session	R-squared:	0.378			
Model:	OLS	Adj. R-squared:	0.366			
Method:	Least Squares	F-statistic:	32.80			
Date:	Mon, 01 Jun 2026	Prob (F-statistic):	4.65e-07			
Time:	18:24:08	Log-Likelihood:	-278.81			
No. Observations:	56	AIC:	561.6			
Df Residuals:	54	BIC:	565.7			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	69.5556	8.785	7.918	0.000	51.944	87.168
Days	1.0586	0.185	5.727	0.000	0.688	1.429
=====						

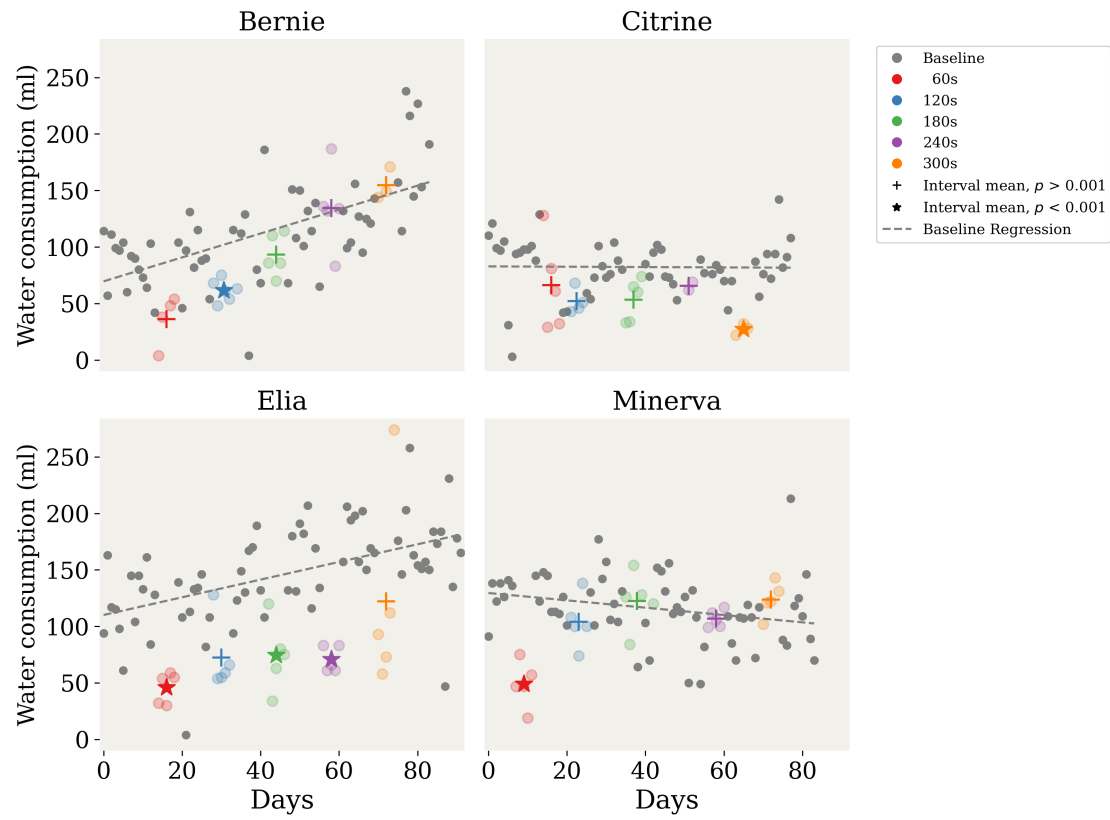


Figure 4: Figure 5. Daily in-session water consumption across baseline and FT intervals.

Omnibus:	1.685	Durbin-Watson:	2.197
Prob(Omnibus):	0.431	Jarque-Bera (JB):	0.944
Skew:	-0.027	Prob(JB):	0.624
Kurtosis:	3.634	Cond. No.	87.3

=====

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

5.1.2 Citrine

OLS Regression Results

```

=====
Dep. Variable:          session    R-squared:                0.000
Model:                  OLS        Adj. R-squared:           -0.019
Method:                 Least Squares    F-statistic:             0.01175
Date:                  Mon, 01 Jun 2026    Prob (F-statistic):      0.914
Time:                  18:24:08          Log-Likelihood:          -243.68
No. Observations:      53              AIC:                     491.4
Df Residuals:          51              BIC:                     495.3
Df Model:              1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	82.7957	6.237	13.276	0.000	70.275	95.316
Days	-0.0150	0.138	-0.108	0.914	-0.293	0.263

=====

Omnibus:	7.717	Durbin-Watson:	1.350
Prob(Omnibus):	0.021	Jarque-Bera (JB):	7.990
Skew:	-0.589	Prob(JB):	0.0184
Kurtosis:	4.494	Cond. No.	83.6

=====

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

5.1.3 Elia

OLS Regression Results

```

=====
Dep. Variable:          session    R-squared:                0.273
Model:                  OLS        Adj. R-squared:           0.261
Method:                 Least Squares    F-statistic:             23.98
Date:                  Mon, 01 Jun 2026    Prob (F-statistic):      6.92e-06
Time:                  18:24:08          Log-Likelihood:          -330.74
No. Observations:      66              AIC:                     665.5
Df Residuals:          64              BIC:                     669.9
Df Model:              1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	110.0446	8.727	12.610	0.000	92.611	127.478
Days	0.7822	0.160	4.897	0.000	0.463	1.101

```
=====
Omnibus:                17.497    Durbin-Watson:                2.039
Prob(Omnibus):           0.000    Jarque-Bera (JB):         29.742
Skew:                   -0.911    Prob(JB):                 3.48e-07
Kurtosis:                5.737    Cond. No.                  105.
=====
```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

5.1.4 Minerva

OLS Regression Results

```
=====
Dep. Variable:          session    R-squared:                0.072
Model:                  OLS        Adj. R-squared:           0.056
Method:                 Least Squares    F-statistic:             4.418
Date:                   Mon, 01 Jun 2026    Prob (F-statistic):       0.0400
Time:                   18:24:08    Log-Likelihood:          -282.10
No. Observations:       59          AIC:                     568.2
Df Residuals:           57          BIC:                     572.4
Df Model:               1
Covariance Type:        nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	129.4832	7.543	17.165	0.000	114.378	144.589
Days	-0.3243	0.154	-2.102	0.040	-0.633	-0.015

```
=====
Omnibus:                10.271    Durbin-Watson:                1.880
Prob(Omnibus):           0.006    Jarque-Bera (JB):         15.433
Skew:                   0.554    Prob(JB):                 0.000445
Kurtosis:                5.247    Cond. No.                  96.5
=====
```

Notes:

[1] Standard Errors assume correct specification of covariance matrix.

5.2 Kolmogorov–Smirnov tests of FT vs. baseline residual distributions

Each row compares the distribution of residuals from the corresponding cat’s baseline regression against the distribution of residuals obtained by applying that same baseline model to the FT interval named in the comparison column. A p -value below 0.001 indicates that the FT-interval residuals differ significantly from baseline residuals — i.e., that the intervention altered in-session water consumption beyond what would be expected from intrinsic baseline variation.

Subject	Comparison	KS statistic	p-value
Bernie	Interval 60s vs Baseline	ks = 0.821	p = 0.004
Bernie	Interval 120s vs Baseline	ks = 0.786	p < 0.001
Bernie	Interval 180s vs Baseline	ks = 0.536	p = 0.094
Bernie	Interval 240s vs Baseline	ks = 0.336	p = 0.574
Bernie	Interval 300s vs Baseline	ks = 0.500	p = 0.371
Citrine	Interval 60s vs Baseline	ks = 0.430	p = 0.281
Citrine	Interval 120s vs Baseline	ks = 0.811	p = 0.005

Subject	Comparison	KS statistic	p-value
Citrine	Interval 180s vs Baseline	ks = 0.698	p = 0.011
Citrine	Interval 240s vs Baseline	ks = 0.811	p = 0.089
Citrine	Interval 300s vs Baseline	ks = 0.962	p < 0.001
Elia	Interval 60s vs Baseline	ks = 0.970	p < 0.001
Elia	Interval 120s vs Baseline	ks = 0.770	p = 0.002
Elia	Interval 180s vs Baseline	ks = 0.818	p < 0.001
Elia	Interval 240s vs Baseline	ks = 0.970	p < 0.001
Elia	Interval 300s vs Baseline	ks = 0.770	p = 0.002
Minerva	Interval 60s vs Baseline	ks = 0.949	p < 0.001
Minerva	Interval 120s vs Baseline	ks = 0.546	p = 0.083
Minerva	Interval 180s vs Baseline	ks = 0.376	p = 0.426
Minerva	Interval 240s vs Baseline	ks = 0.407	p = 0.339
Minerva	Interval 300s vs Baseline	ks = 0.563	p = 0.067